

Table 1. *RBC-5wk* general information

Building	Location	Real/simulated	Control strategy	Number of faults	Number of features	Sampling rate	Data duration
Medium-sized office building	Tuscaloosa, AL, USA	Simulated	RBC	21	114	5-min	5 weeks

Table 2. Details of *RBC-5wk* fault scenarios. Faults were injected during occupied mode duration, which was weekdays (Mon-Fri) from 7 a.m. to 7 p.m.

Fault abbreviation (CSV file name)	Fault type	Fault intensity	Method of fault imposition
CooCoiValStuck_0	Cooling coil valve stuck	0%	Defining the leakage ratio as: $l = \frac{\kappa v(y=0)}{\kappa v(y=1)}$, where $\kappa v(y = 0)$ and $\kappa v(y = 1)$ are the flow coefficient at the fully closed and fully open positions, respectively.
CooCoiValStuck_5		5%	
CooCoiValStuck_15		15%	
CooCoiValStuck_65		65%	
CooCoiValStuck_100		100%	
OADamStuck_0	Outdoor air damper stuck	0%	
OADamStuck_5		5%	
OADamStuck_15		15%	
OADamStuck_45		45%	
OADamStuck_55		55%	
OADamStuck_65		65%	
OADamStuck_100		100%	
SupDucLea_20	Supply duct leakage	20%	Implementing the mathematical expression as: $\dot{m}_{EF} + \dot{m}_L = k\sqrt{\Delta p}$, where $\dot{m}_{EF}$ is the normal operation mass flow rate, $\dot{m}_L$ is the leakage mass flow rate, k is the friction factor, and Δp is the pressure drop.
SupDucLea_45		45%	
SupDucLea_55		55%	
TSup_p2	AHU supply air temperature sensor bias	+2 °K	The output (sensed variable) of the original sensor model is overwritten by the faulty value.
TSup_m2		-2 °K	
Vout_p30	Outdoor air flow rate sensor scale error	+30%	

Vout_m30		-30%	
RelPreCHW_p10000	Chilled water differential pressure sensor bias	+10,000 Pa	
RelPreCHW_m10000		-10,000 Pa	

Additional information

- All simulations began at $t = 15,724,800$ sec. (corresponding to the start of the 183rd day of the year, July 2nd) and ended at $t = 18,748,800$ sec. (the end of the 217th day of the year, August 5th), covering a total duration of five weeks in each CSV file. Faults were introduced starting at $t = 16,354,800$ sec. (7:00 a.m. on the 190th day, July 9th) and continued until $t = 18,730,800$ sec. (7:00 p.m. on the 217th day, August 5th). Consequently, each fault simulation spans four weeks, with the first week in each fault CSV file being fault-free.
- Please note, since this dataset represents a simulated building, the baseline (fault-free) data was intentionally generated with the same timestamp as the fault data. This was achieved by applying identical outdoor conditions (e.g., outdoor air humidity and temperature), occupancy load, time of day, simulation period, and other related parameters when generating the dataset from the simulated model. This approach allows researchers to compare the corresponding baseline and fault/attack datasets without confounding variations from external factors.
- Fig. 1 shows the schematic of the corresponding AHU system. The system is a single-duct multi-zone VAV AHU connected with five VAV terminal boxes that serve five zones (four exterior and one interior zone) on a single floor.

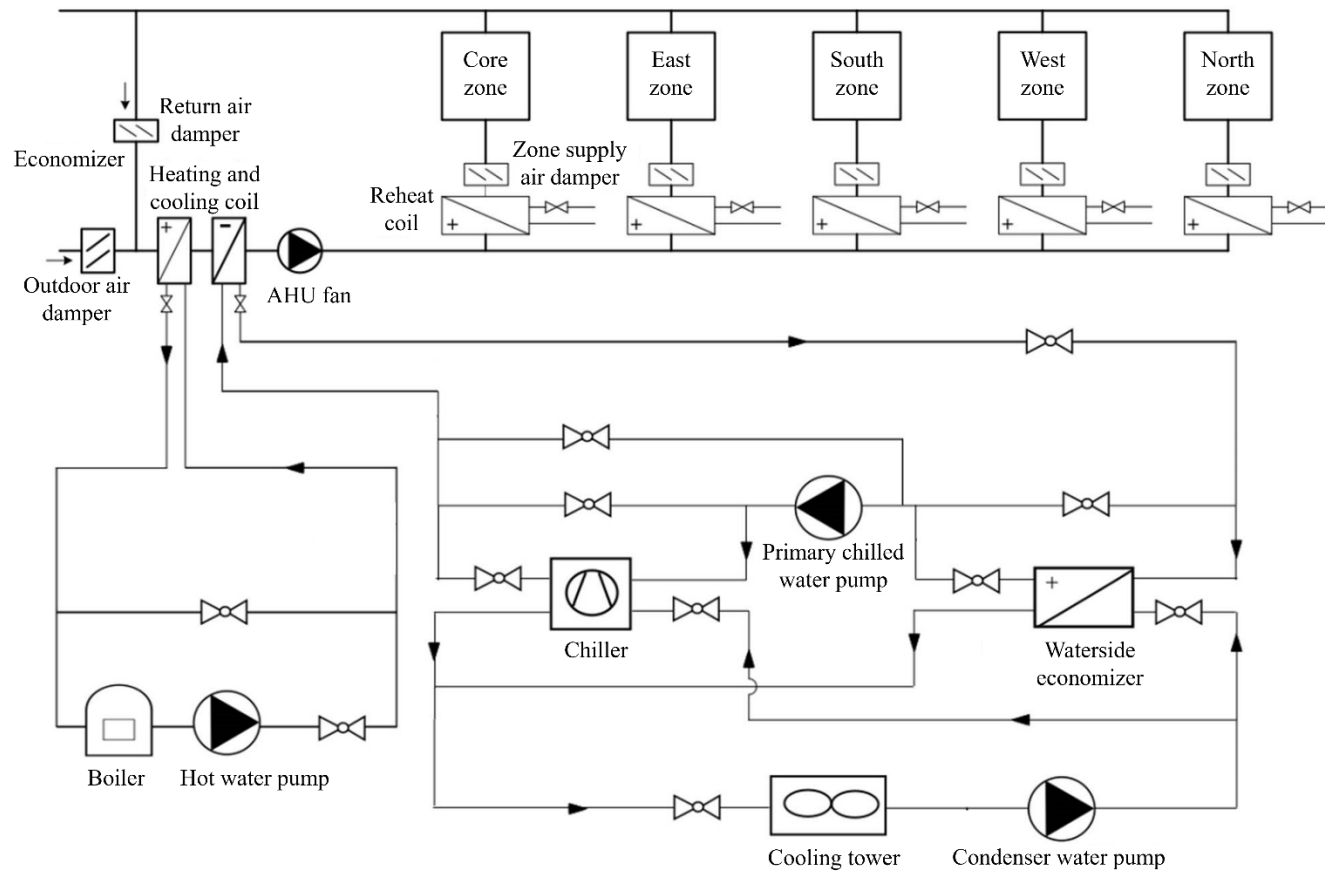


Fig. 1. Schematic of the AHU diagram, used to generate the dataset.